

ORIE 5530: Assignment 1

Due on Friday Sep 11, 11:59pm (EST)

Notes:

- All questions in this homework must be individual work. You are allowed to discuss the assignment with others but please mention in your write-up if you have discussed the solution with someone.
- Problem 2 and 4 contain coding questions. You should also submit your programming code (Python, R, Matlab, etc) along with the answers. You are allowed to use any programming language, though we strongly recommend Python.

Problem 1 (Events, 24 points) Let B, H, S be three events (say, B is “Biden wins the elections”, H is “Democrats control the house after the elections” and S is “Democrats control the senate after the elections”). Find expressions for the events:

(a) only H occurs

$$\boxed{Pr(H) - [(Pr(B \cap H) + Pr(H \cap S) - Pr(B \cap H \cap S))]},$$

(b) both B and H but not S occur

$$\boxed{Pr(B \cap H) - Pr(B \cap H \cap S)},$$

(c) at least one event occurs

$$\boxed{Pr(B \cup H \cup S)},$$

(d) at least two events occur

$$\boxed{Pr(H \cap B) + Pr(B \cap S) + Pr(H \cap S) - 2 * Pr(H \cap B \cap S)},$$

(e) all three events occur

$$\boxed{Pr(B \cap H \cap S)},$$

(f) none occurs

$$\boxed{1 - Pr(B \cup H \cup S)},$$

(g) at most one occurs

$$\boxed{Pr(B \cup H \cup S) - [Pr(B \cap H) + Pr(H \cap S) + Pr(S \cap B) - 2 * Pr(H \cap B \cap S)]}$$

(h) at most two occur

$$\boxed{Pr(B \cup H \cup S) - Pr(H \cap B \cap S)}.$$

Problem 2 (Counting, 33 points) You have N cents and m possible coin denominations $\{S_1, \dots, S_m\}$ (say $S_1 = 1$ cent, $S_2 = 2$ cents, etc.). We want to know in how many ways we can break the N cents into combinations of coins. For example, let $S_1 = 1$, $S_2 = 5$ and $S_3 = 10$, and $N = 10$. Then, we have 4 options: (i) 1 coin of 10 cents, (ii) two coins of 5 cents each (iii) 5 coins of 1 cent and 1 coin of 5 cents (iv) 10 coins of one cent. If you have only $S_1 = 1$ and $S_2 = 5$ (i.e., you are not allowed to use the 10 cents coin) then you have only three options. In this example 75% of the outcomes do not use 10 cents.

Python code for Problem 2

```
import matplotlib.pyplot as plt
import numpy as np

#Problem 2 - Counting, 33 points
## (a) - 15 points
def C(N, m):
    # base cases
    if N == 0:
        return 1                # if N has reached 0, then this is
        ↪ valid
    if len(m) == 0:
        return 0                # if m is empty then we have run out
        ↪ of denominations
    if N < 0:
        return 0                # if N is negative then we have
        ↪ overshoot the target

    # setup
    num_coins = len(m)          # get the length of m
    largest = m[num_coins - 1]  # get the largest denomination

    # recursive cases
    without_Sm = C(N, m[0 : num_coins-1]) # do the withouts first (exclude
    ↪ largest denomination entirely but not change target N unchanged). this
    ↪ will run until base case is reached.
    with_Sm = C(N - largest, m)           # use one coin of the largest
    ↪ denomination, still allowed to use it again, so m unchanged

    return without_Sm + with_Sm

#test cases
print (C(10, [1,5,10]))
# 1. C(10, [1,5,10])
# -> without_Sm[1] = C(10, [1,5]) = without_Sm[2] + with_Sm[2] = 3
# 2. C(10, [1,5])
# -> without_Sm[2] = C(10, [1]) = without_Sm[3] + with_Sm[3] = 1
# 3. C(10, [1])
# -> without_Sm[3] = C(10, []) = 0
# -> with_Sm[3] = C(9, [1]) = ... = 1
```

```

#         -> with_Sm[2] = (5, [1,5]) = without_Sm[4] + with_Sm[4] = 2
#         4. (5, [1,5])
#         -> without_Sm[4] = (5, [1]) = ... = 1
#         -> with_Sm[4] = (0, [1,5]) = 1
#         -> with_Sm[1] = C(0, [1,5,10]) = 1
# Final Output = without_Sm[1] + with_Sm[1] = 4

print (C(2, [1, 2]))
# 1. C(2, [1,2]).
# -> without_Sm[1] = C(2, [1]) = without_Sm[2] + with_Sm[2] = 1
# 2. C(2, [1]).
# -> without_Sm[2] = C(2, []) = 0
# -> with_Sm[2] = C(1, [1]) = without_Sm[3] + with_Sm[3] = 1
# 3. C[1, [1]]
# -> without_Sm[3] = C[1, []] = 0
# -> with_Sm[3] = C[0, [1]] = 1
# -> with_Sm[1]: Call C(0, [1, 2]) = 1
# Final Output = without_Sm[1] + with_Sm[1] = 2

## (b) - 5 points
print(C(213, [1,5,10,25]))
### 1670 ways

## (c) - 5 points
### create wrapper function that only calls it once
def C_firstround(N, m):
    num_coins = len(m)
    largest = m[num_coins - 1]
    without_Sm = C(N, m[0 : num_coins-1])
    with_Sm = C(N - largest, m)
    return without_Sm, with_Sm, without_Sm / (without_Sm + with_Sm)

print(C_firstround(213, [1,5,10,25]))
# -> without_Sm = C(213, [1,5,10]) = 484
# -> with_Sm = C(213, [1,5,10,25]) = 1186
# 0.2898203592814371

## (d) -
x_val = list()
y_val = list()
for i in range(25, 500, 3):
    x_val.append(i)
    y_val.append(C_firstround(i, [1,5,10,25])[2])

# print (x_val)
# print (y_val)

plt.scatter(x_val, y_val)

```

```
plt.xlabel("N")
plt.ylabel("Fraction of Counts Using Only [1,5,10]")
plt.show()
```

Written responses for Problem 2

- (a) (15 points) Write a recursion of $C(N, m)$ the number of ways to “build” N cents if you have the denominations $\{S_1, \dots, S_m\}$. Code the recursion.

Hint 1: Split into combinations that use the denomination S_m and those that do not.

Hint 2: Write $C(N, m)$ as a sum $C(x, m) + C(N, y)$, where you have to carefully choose the values x and y such that $x \leq N$ and $y \leq m$.

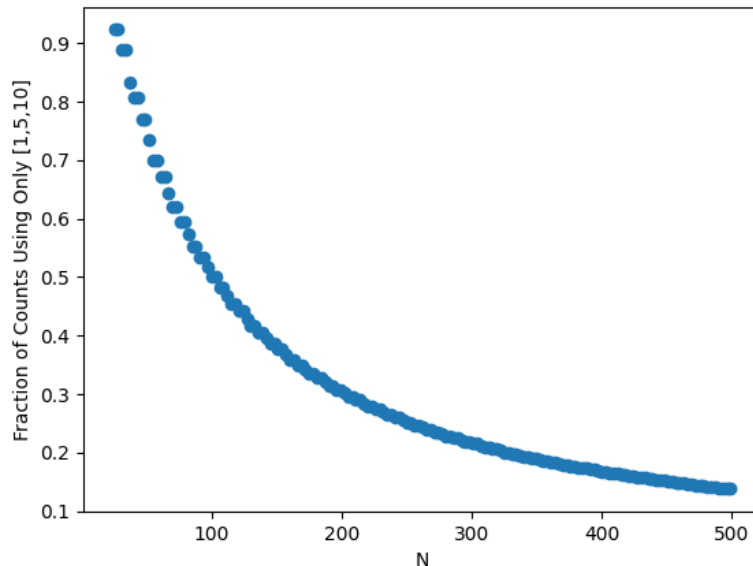
- (b) (5 points) With $S_1 = 1, S_2 = 5, S_3 = 10$, and $S_4 = 25$, how many ways there are to change $N = 213$?

1670 This is the output of $(C(213, [1,5,10,25]))$.

- (c) (5 points) What fraction of these uses only 1, 5 and 10 cents denominations?

0.2898203592814371 Without $S_m = C(213, [1,5,10]) = 484$ while With $S_m = C(213, [1,5,10,25]) = 1186$ which is $\frac{484}{1670} = 0.2898203592814371$.

- (d) (8 points) Repeat the same for multiple values of N and plot the result to the previous question as a function of N ?



Problem 3 (Conditional probabilities, 20 points). Consider two boxes A and B . Box A contains two balls: one red and one white ball. Box B contains three balls: two red and one white ball. A box is selected at random (that is, you select box A with probability 0.5) and a ball is drawn at random from the selected box.

- (a) (10 points) What is the probability that the ball is red?

$\boxed{\frac{7}{12}}$ because $Pr(R) = Pr(R|A) * Pr(A) + Pr(R|B) * Pr(B)$ given $Pr(R)$ is probability that ball is red, $Pr(A)$ is probability that Box A is chosen, and $Pr(B)$ is probability that Box B is chosen. This gives $\frac{1}{2} * \frac{1}{2} + \frac{2}{3} * \frac{1}{2} = \frac{7}{12}$

- (b) (10 points) Say you can see the final outcome (the color of the selected ball) but you do not know which box it was selected from. What is the probability that box A was the one selected given that the ball is white?

$\boxed{\frac{3}{5}}$ because $Pr(A|W) = \frac{Pr(W \cap A)}{Pr(W)} = \frac{\frac{1}{2} * \frac{1}{2}}{\frac{1}{2} * \frac{1}{2} + \frac{1}{3} * \frac{1}{2}} = \frac{3}{5}$.

Problem 4 (Covid19, 23 points). Consider the coronavirus data posted by NYC Department of Health and Mental Hygiene at <https://github.com/nychealth/coronavirus-data/blob/master/trends/cases-by-day.csv>. The file records the number of new cases every day in each boroughs of NYC. Column `date_of_interest` corresponds to the date and `CASE_COUNT` corresponds to the total number of infected cases in NYC on each day. Column `MN_CASE_COUNT` records the number of infected cases in Manhattan on each day. Manhattan's population is 1,694,263 and the total population in NYC is 8,804,190. Analyze the data and answer the following questions:

Python code for Problem 4

```
import array
import numpy
import csv

#Problem 4 - Covid19, 23 points
## (a) - 11 points
# Pr(Infected | Lived in Manhattan) = Pr (Lived in Manhattan and was Infected) /
#   ↳ Pr (Lived in Manhattan)
# pseudocode
# imports
# import numpy
# import csv file cases-by-day.csv
#
# set variables
# pop_nyc = 8804190
# pop_mn = 1694263
# cum_case_count = 0
# cum_mn_case_count = 0
# i = 0
#
# loop
# while date_of_interest[i] >= "01/01/2021" and date_of_interest[i] <=
#   ↳ "01/31/2021":
#   cum_case_count = cum_case_count + case_count[i]
#   cum_mn_case_count = cum_mn_case_count + cum_mn_case_count[i]
#   i++
```

```

# print(date_of_interest[i], cum_case_count, cum_mn_case_count)
#return

pop_nyc = 8804190
pop_mn = 1694263

#open csv
with open("./cases-by-day.csv", newline="") as f:
    reader = csv.DictReader(f)
    rows = list(reader)

def case_count(x,y):
    #set variables
    cum_case_count = 0
    cum_mn_case_count = 0

    #find date indices
    dates = []
    for row in rows:
        dates.append(row["date_of_interest"])

    start = dates.index(x)
    end = dates.index(y)

    #loop
    for i in range(start, end + 1):
        cum_case_count += int(rows[i]["CASE_COUNT"])
        cum_mn_case_count += int(rows[i]["MN_CASE_COUNT"])

    print(f"dates_of_interest: {dates[start]}:{dates[end]}, cum_case_count:
    ↪ {cum_case_count}, cum_mn_case_count: {cum_mn_case_count}")
    return (dates[start],dates[end], cum_case_count, cum_mn_case_count)

# calculating a
tup_case_count_jan2021 = case_count("01/01/2021", "01/31/2021")

cum_mn_case_count_jan2021 = tup_case_count_jan2021[3]
p_mn_and_i_jan2021 = cum_mn_case_count_jan2021/pop_nyc
p_mn = pop_mn / pop_nyc

p_i_given_mn_jan2021 = p_mn_and_i_jan2021 / p_mn
print(p_i_given_mn_jan2021) #0.011410861241731655

## (b) - 6 points
cum_case_count_jan2021 = tup_case_count_jan2021[2]
p_mn_and_i_jan2021 = cum_mn_case_count_jan2021/pop_nyc
p_i_jan2021 = cum_case_count_jan2021 / pop_nyc

```

```

p_mn_given_i_jan2021 = p_mn_and_i_jan2021 / p_i_jan2021
print(p_mn_given_i_jan2021) #0.1395048454716668

## (c) - 6 points
p_i2024_given_i2021 = 0.05
p_i2024_given_noti2021 = 0.20
p_i2021_given_i2024 = (p_i2024_given_i2021 * p_i_jan2021) / (p_i2024_given_i2021 *
↪ p_i_jan2021 + p_i2024_given_noti2021 * (1-p_i_jan2021))
print(p_i2021_given_i2024) #0.003982154426490655

```

Written responses for Problem 4

- (a) (11 points) In January 2021, suppose an NYC resident lived in Manhattan, what is the probability that he was infected?

0.011410861241731655 because $Pr(I|M) = \frac{Pr(I \cap M)}{Pr(M)}$

- (b) (6 points) In January 2021, suppose an NYC resident was infected, what is the probability that he lived in Manhattan?

0.1395048454716668 because $Pr(M|I) = \frac{Pr(I \cap M)}{Pr(I)}$

- (c) (6 points) Assume the following:

- If an NYC resident was infected in January 2021, then the probability that they get infected in January 2024 is 5%.
- If an NYC resident was NOT infected in January 2021, then the probability that they get infected in January 2024 is 20%.

Question: If an NYC resident gets infected in January 2024, what is the probability that they were also infected in January 2021?

0.003982154426490655 because $Pr(I2021|I2024) = \frac{Pr(I2024|I2021)*Pr(I2021)}{Pr(I2024|I2021)*Pr(I2021)+Pr(I2024|I2021^c)*Pr(I2021^c)}$ and we know $Pr(I2024|I2021) = 0.05$, $Pr(I2021) = \frac{138583}{8804190}$ through analysis, $Pr(I2024|I2021^c) = 0.20$, and $Pr(I2021^c) = 1 - \frac{138583}{8804190}$, then $Pr(I2021|I2024) = 0.003982154426490655$.